

Roe Notes

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1 Riemannian Geometry

1.1 Connections

Let M be a smooth manifold, and let V be a vector bundle on M .

Definition 1.1. A *Connection* on V is a linear map

$$\nabla : C^\infty(TM) \otimes C^\infty(V) \rightarrow C^\infty(V)$$

which assigns to a vector field X and a section Y of V a new section $\nabla_X Y$ of V , such that for any smooth function f on M ,

1. $\nabla_{fX} Y = f \nabla_X Y$
2. $\nabla_X fY = f \nabla_X Y + (X.f)Y$,

where $X.f$ denotes the lie derivative of f along X .

Condition (1.) tells us that $X \mapsto \nabla_X Y$ is a homomorphism of modules over $C^\infty(M)$. In other words, a connection is tensorial in $C^\infty(TM)$. This means that we can regard a connection as a map

$$C^\infty(V) \rightarrow \Omega^1(V) := C^\infty(T^*M \otimes V),$$

the space of V valued forms on M .

In a local trivialization, an example of a connection is the Lie derivative. Thus many local connections exist.

Moreover, the difference of two connections is local in both X and Y , making it an $\text{End}(V)$ valued one-form. Thus, we can use partitions of unity to piece together local connections into a global one. So connections exist and are plentiful.

From the above discussion, we deduce that in a local trivialization,

$$\nabla_i = \frac{\partial}{\partial x_i} + \Gamma_i,$$

where $\nabla_i = \nabla_{\frac{\partial}{\partial x_i}}$, and Γ_i is a smooth section of $\text{End}(V)$. The Γ_i are called the Christoffel symbols.

Remark 1.1. We can also define connections using *paralell transport*. Let γ be a smooth curve in M . The differential equation $\nabla_{\dot{\gamma}} Y = 0$ is a first order ODE on sections Y of V along γ which has a unique solution for a given initial value. One says that Y is paralell along γ , or that Y has been obtained from its initial value by paralell transport.

Definition 1.2. The *curvature operator* K of a connection ∇ on V is defined as follows: if X and Y are vector fields and Z is a section of V , then

$$K(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Lemma 1.1. *The curvature $F(X, Y)Z$ at a point $m \in M$ depends only on the values of X, Y , and Z at the point m .*

Proof: Choose some local coordinate patch and a trivialization of V . Then we can expand

$$\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z = \sum_{i,j} \left[X^j \frac{\partial Y^i}{\partial x_j} - Y^j \frac{\partial X^i}{\partial x_j} \right] \frac{\partial Z}{\partial x_i} + [X^j Y^i - Y^j X^i] \frac{\partial^2 Z}{\partial x_j \partial x_i} \quad (1)$$

$$+ [X^j Y^i - Y^j X^i] \left(\frac{\partial \Gamma_j(Z)}{\partial x_i} + \frac{\partial \Gamma_i(Z)}{\partial x_j} \right) \quad (2)$$

$$+ \left[X^j \frac{\partial Y^i}{\partial x_j} - Y^j \frac{\partial X^i}{\partial x_j} \right] (\Gamma_i(Z)) + [X^j Y^i - Y^j X^i] (\Gamma_j \Gamma_i(Z)). \quad (3)$$

Since mixed partials commute, the second term in (1) vanishes. By symmetry, the term (2) vanishes as well, leaving only

$$\begin{aligned} \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z &= \sum_{i,j} \left[X^j \frac{\partial Y^i}{\partial x_j} - Y^j \frac{\partial X^i}{\partial x_j} \right] \left(\frac{\partial Z}{\partial x_i} + \Gamma_i(Z) \right) \\ &\quad + [X^j Y^i - Y^j X^i] (\Gamma_j \Gamma_i(Z)) \\ &= \nabla_{[X, Y]} Z + [X^j Y^i - Y^j X^i] \Gamma_j \Gamma_i(Z). \end{aligned}$$

From here the lemma follows immediately. $\square$

Thus K is induced by a bundle map $TM \otimes TM \rightarrow \text{End}(V)$, and moreover since $K(X, Y)$ is antisymmetric in X and Y , we may regard K as a 2-form with values in $\text{End}(V)$. In terms of local coordinates, we define the curvature 2-form to be the $\text{End}(V)$ -valued 2-form

$$K = \sum_{i < j} K(\partial_i, \partial_j) dx^i \wedge dx^j,$$

where $K(\partial_i, \partial_j)$ takes values in $\text{End}(V)$.

Now we consider connections on the tangent bundle TM . Let x^i be local coorinates with a corresponding vector fields ∂_i . Then we can express the endomorphisms Γ_i of remark (1.1) as matrices:

$$\nabla_i(\partial_j) = \sum_k \Gamma_{ij}^k \partial_k.$$

These symbols completely determine the connection.

Definition 1.3. A connection ∇ on TM is called *symmetric* if for any two vector fields X and Y on TM , we have

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

1.2 Riemannian Geometry

Suppose M is a Riemannian manifold.

Definition 1.4. We say ∇ is *compatible with the metric* on M if for any three vector fields X , Y_1 , and Y_2 , we have

$$(\nabla_X Y_1, Y_2) + (Y_1, \nabla_X Y_2) = X \cdot (Y_1, Y_2).$$

Theorem 1.1 (Levi-Civita). *A Riemannian manifold possesses a unique connection that is symmetric and compatible with the metric.*

Proof: Work in local coordinates. Represent the metric by a matrix of smooth functions $g_{jk} = (\partial_j, \partial_k)$. The symmetry condition implies

$$(\nabla_i \partial_j, \partial_k) + (\partial_j, \nabla_i \partial_k) = \partial_i (\partial_j, \partial_k),$$

or in other words,

$$\partial_i g_{jk} = \sum_a \left(\Gamma_{ij}^a g_{ak} + \Gamma_{ik}^a g_{aj} \right).$$

Moreover, we can permute indices to get

$$\partial_j g_{ki} = \sum_a \left(\Gamma_{jk}^a g_{ai} + \Gamma_{ji}^a g_{ak} \right) \quad \partial_k g_{ij} = \sum_a \left(\Gamma_{ki}^a g_{aj} + \Gamma_{jk}^a g_{ai} \right).$$

Combining these and using symmetry, we see that

$$\sum_a \Gamma_{ij}^a g_{ak} = \frac{1}{2} (\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}).$$

This determines the Γ 's uniquely, since g is invertible. □

Since the metric determines the Levi-Civita connection, it also uniquely determines the curvature.

Definition 1.5. The curvature of the Levi-Civita connection on a Riemannian manifold is called the *Riemann curvature operator* and is denoted by R . In components relative to a frame $\{e_i\}$ we write

$$R(e_j, e_k)e_l = \sum_i R_{ljk}^i e_i.$$

We also define the 4-covariant version of the Riemannian curvature tensor defined by

$$R_{iljk} = (R(e_j, e_k)e_l, e_i).$$

Proposition 1.1. *The Riemann curvature operator has the following properties:*

- (i) $R(X, Y)Z + R(Y, X)Z = 0$, or $R_{ljk}^i + R_{lkj}^i = 0$.
- (ii) $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$, or $R_{ljk}^i + R_{jkl}^i + R_{klj}^i = 0$.
- (iii) $(R(X, Y)Z, W) + (R(X, Y)W, Z) = 0$, or $R_{iljk} + R_{lijk} = 0$.
- (iv) $(R(X, Y)Z, W) = (R(Z, W)X, Y)$, or $R_{iljk} = R_{jkil}$.

The only non-trivial contraction of the Riemann curvature tensor is called the *Ricci curvature tensor*, given by

$$\text{Ric}(Y, Z) \mapsto \text{tr}(X \mapsto R(X, Y)Z).$$

In components, $\text{Ric}_{ab} = \sum_i R_{aib}^i$.

We can raise one index of the Ricci curvature tensor to get the Ricci curvature operator, and taking its trace we obtain the scalar curvature,

$$\kappa = g^{ab} \text{Ric}_{ab}.$$

Definition 1.6. We say a curve γ is a *geodesic* if $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$, or in other words if $\dot{\gamma}$ is parallel along γ .

The geodesic equation has a unique solution given initial values $\gamma(0)$ and $\dot{\gamma}(0)$. Then the exponential map

$$\exp : U \rightarrow M$$

can be defined at any $m \in M$ on some open star-shaped subset $U \subset T_m M$ by sending a vector $v \in T_m M$ to the value at $t = 1$ of the unique geodesic with starting value m and velocity v . By the inverse function theorem, $\exp$ is a diffeomorphism from a neighborhood of zero in $T_m M$ to a neighborhood of m in M . Choosing an orthonormal basis for $T_m M$ gives a coordinate system called a *geodesic coordinate system* at m .

Proposition 1.2. *At the origin of a geodesic coordinate system, the Christoffel symbols all vanish.*

Proof: It is enough to show that $\nabla_i \partial_j$ vanishes at the origin. By symmetry of the connection, it is also enough to show $\nabla_X X = 0$ for all constant vector fields X .

But in a geodesic coordinate system, the geodesics are radial lines from the origin. Moreover, X clearly points along these geodesics and has constant length, so $\nabla_X X = 0$. $\square$

1.3 Differential forms

We let $\Omega^p = \wedge^p V^*$ denote the space of p -forms. We can identify p -forms with antisymmetric tensors by writing

$$\alpha = \sum_{i_1 < \dots < i_p} A_{i_1, \dots, i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p} = \frac{1}{p!} \sum_{i_1, \dots, i_p} A_{i_1, \dots, i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}.$$

We first extend the metric to the space of contravariant tensors.

Let $(\cdot, \cdot)$ be an inner product on V , and fix a basis $\{e^i\}$ of V , with corresponding dual basis $\{e_i\}$. We have the matrix $g_{ij} = (e_i, e_j)$ and its inverse g^{ij} .

Then we have the lowering and raising of indices maps,

$$\flat : e^i \mapsto g_{ij} e_j$$

$$\sharp : e_i \mapsto g^{ij} e^j.$$

These extend to tensors in the natural way.

Using the metric, we can define one on the space of $(0, p)$ tensors by the formula

$$(\alpha, \beta) = \frac{1}{k!} \sum_{i_1, \dots, i_p, j_1, \dots, j_p} g^{i_1 j_1} \dots g^{i_p j_p} A_{i_1, \dots, i_p} B_{j_1, \dots, j_p}.$$

In an oriented coordinate system, we define the volume form $\text{vol} = \sqrt{g} dx^1 \wedge \cdots \wedge dx^n$, where $g = \det g_{ij}$. The normalization constant in the inner product formula ensures that $(\text{vol}, \text{vol}) = 1$:

$$\begin{aligned} (\text{vol}, \text{vol}) &= \frac{1}{n!} \sum_{i_1, \dots, i_n, j_1, \dots, j_n} g^{i_1, j_1} \cdots g^{i_n, j_n} \delta_{1, \dots, n}^{i_1, \dots, i_n} \delta_{1, \dots, n}^{j_1, \dots, j_n} \\ &= \frac{1}{n!} g \sum_{j_1, \dots, j_n} \det(g^{ij}) = 1. \end{aligned}$$

We also have formulas for the wedge product and exterior derivative of forms in terms of the antisymmetric tensors associated to them. If C corresponds to $\alpha \wedge \beta$, then

$$C_{k_1, \dots, k_{p+q}} = \frac{1}{p!q!} \sum_{i_1, \dots, i_p, j_1, \dots, j_q} \delta_{k_1, \dots, k_{p+q}}^{i_1, \dots, i_p, j_1, \dots, j_q}.$$

and if $D \sim d\alpha$, then

$$D_{k_1, \dots, k_{p+1}} = \frac{1}{p!} \sum_{i_1, \dots, i_p} \delta_{k_1, \dots, k_{p+1}}^{j, i_1, \dots, i_p} \frac{\partial A_{i_1, \dots, i_p}}{\partial x_j}.$$

Definition 1.7. Let α be a k -form. Define the *Hodge Star* $*\alpha$ to be the unique $(n - k)$ form such that for all k -forms β ,

$$(\alpha, \beta) \text{vol} = \beta \wedge *\alpha.$$

This operation is linear and we have $**\alpha = (-1)^{kn+k}\alpha$

Definition 1.8. If α is a k -form, define

$$d^*\alpha = (-1)^{nk+n+1} * d * \alpha.$$

Then $d^*\alpha$ is a $(k - 1)$ form, and $(d^*)^2 = 0$.

The importance of d^* is that it is the adjoint of d , once we define a global inner product on forms. We define

$$\langle \alpha, \beta \rangle = \int_M (\alpha, \beta) \text{vol} = \int_M \beta \wedge *\alpha = \int_M$$

1.4 Exercises

1.28 Prove Cartan's formula for the exterior derivative of a p -form.

Proof: See homework either 6 or 7 from DG.

1.29 Prove the second Bianchi Identity,

$$\nabla R(X, Y, Z, V, W) + \nabla R(X, Y, V, W, Z) + \nabla R(X, Y, W, Z, V) = 0.$$

Proof: Since the objects are tensors, it suffices to prove the identity at an arbitrary point on M . Fix a geodesic coordinate system about this point so that brackets and $\nabla_i e_j$ vanishes at the origin.

Then follow the steps in Do Carmo with the simplified coordinate system, then the result is clear.

- 1.30 Let x^i be a geodesic coordinate system on a manifold M , and let r denote the Riemannian distance from the point x to the origin. Prove that

$$\sum_{i,j} g_{ij}(x) x^i x^j = \sum_{i,j} g^{ij}(x) x^i x^j = r^2.$$

Proof: This problem is straightforward using the fact that in a geodesic coordinate system, the geodesics are just radial lines from the origin. The distance from the origin to x is the infimum over all geodesics connecting these points of the length of these curves. But in this case, there is only one radial line, so its length is the inf.

Moreover, the velocity of the geodesic at the origin at $t = 0$ and at x at $t = 1$ is just the magnitude of the vector x , viewed as a subset of $T_0 M$.

- 1.31 Let α be a k -form on an n -dimensional oriented Riemannian manifold M represented in local coordinates by an antisymmetric tensor A . Show that

$$*\alpha = \sum_{i_1, \dots, i_k, j_1, \dots, j_{n-k}} \frac{1}{(n-k)!k!} \delta_{i_1, \dots, i_k, j_1, \dots, j_{n-k}}^{1, \dots, n} \sqrt{g} A^{i_1, \dots, i_k} dx^{j_1} \wedge \dots \wedge dx^{j_{n-k}}.$$

Proof: We utilize the antisymmetric tensor representation formula for the wedge product:

$$C_{k_1, \dots, k_{p+q}} = \frac{1}{p!q!} \sum_{i_1, \dots, i_p, j_1, \dots, j_q} \delta_{k_1, \dots, k_{p+q}}^{i_1, \dots, i_p, j_1, \dots, j_q},$$

where C is the tensor representing the wedge of a p -form and a q -form.

We also have a formula for the inner product of k forms:

$$(\alpha, \beta) = \frac{1}{k!} \sum_{i_1, \dots, i_p, j_1, \dots, j_p} g^{i_1, j_1} \dots g^{i_p, j_p} A_{i_1, \dots, i_p} B_{i_1, \dots, i_p}.$$

By definition of $*\alpha$, for all k -forms β , we have

$$(\alpha, \beta) \text{vol} = \beta \wedge *\alpha.$$

In other words, if we write $*A$ and B for the antisymmetric tensors corresponding to $*\alpha$ and β , this means

$$\sqrt{g} \frac{1}{k!} \sum_{|I|=|J|=k} g^{I,J} A_I B_J = \frac{1}{(n-k)!k!} \sum_{|I|=k, |J|=n-k} \delta_{1, \dots, n}^{I,J} B_I (*A)_J.$$

We start from the left. Notice this is just the raising of the indices of A :

$$A^J = \sum_{|I|=|J|=k} g^{I,J} A_I.$$

Then we have

$$(\alpha, \beta) \text{vol} = \sqrt{g} \frac{1}{k!} \sum_{|J|=k} A^J B_J.$$

Now we need a trick with the generalized Kronoker-delta symbols. For k -tuples I, J and an $n-k$ -tuple K , we have

$$\delta_{1,\dots,n}^{IK} \delta_{1,\dots,n}^{JK} \delta_J^I = 1.$$

To see why, consider either when I and J are opposite sign, or the same.

Then we can write

$$\begin{aligned} \sqrt{g} \frac{1}{k} \sum_{|J|=k} A^J B_J &= \frac{1}{(n-k)!k!} \sqrt{g} \frac{1}{k!} \sum_{|I|=|J|=k, |K|=n-k} \delta_{1,\dots,n}^{IK} \delta_{1,\dots,n}^{JK} \delta_J^I B_J A^I \\ &= \frac{1}{(n-k)!k!} \sqrt{g} \frac{1}{k!} \sum_{|I|=|J|=k, |K|=n-k} \delta_{1,\dots,n}^{IK} \delta_{1,\dots,n}^{JK} B_J A^I \\ &= \frac{1}{(n-k)!k!} \sum_{|J|=k, |K|=n-k} \delta_{1,\dots,n}^{IJ} B_I \left(\sqrt{g} \frac{1}{k!} \sum_{|I|=k} \delta_{1,\dots,n}^{IK} A^I \right) \\ &= B \wedge *A, \end{aligned}$$

where we write $B \wedge *A$ to be the tensor for $\beta \wedge *\alpha$. In other words, the matrix in parenthesis with parameter K is $*A$.

test again more